GCSE CHEMISTRY

Higher Tier – Paper 1 style

Topic 4.2: Bonding, structure and properties of matter

Practice questions in the style of AQA GCSE Chemistry (8462). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.

Time allowed	About 155 minutes (roughly 1 minute per mark)
Materials	For this paper you must have: a ruler, a pencil, a calculator and the Periodic Table at the back of the question section.
Instructions	• Use black ink or a black ball-point pen (pencil for diagrams and graphs). • Answer **all** questions. • Answer the questions in the spaces provided. • Do all rough work on a separate sheet. • Cross through any work you do not want to be marked.
Information	• The maximum mark for this paper is **149**. • The marks for questions are shown in brackets. • You are allowed to use a calculator. • You are reminded of the need for good English and clear presentation in your answers. • Questions that ask you to *describe* or *explain* are marked for the science you include, so give each point its own sentence.

Name: _____ Date: _____ Mark: _____ / 149

0 1 This question is about chemical bonding.

0 1 . 1 Tick **one** box in each row to show the type of bonding in each substance. **[3 marks]**

Substance	Ionic	Covalent	Metallic
sodium chloride			
carbon dioxide			
copper			
magnesium oxide			
methane			
iron			

0 1 . 2 Complete the sentences using words from the box. **[2 marks]**

ionic covalent metallic molecular

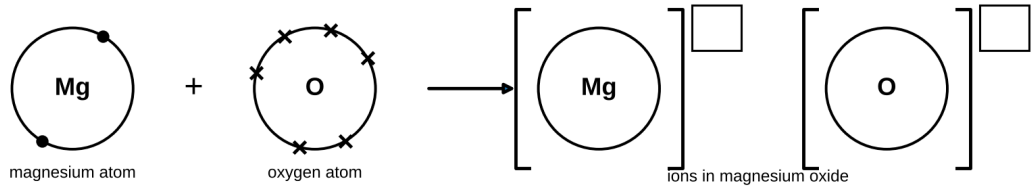
Compounds formed when a metal reacts with a non-metal are _____ compounds.
Compounds formed when non-metals react with other non-metals have _____ bonds.

0 2 This question is about ionic bonding.

0 2 . 1 Sodium (electronic structure 2,8,1) reacts with chlorine (electronic structure 2,8,7) to form sodium chloride. **[4 marks]**
Describe, in terms of electrons, what happens when sodium reacts with chlorine.

0 2 . 2 The diagram shows how magnesium and oxygen atoms form magnesium oxide. Dots (•) show electrons from magnesium. Crosses (×) show electrons from oxygen. **[3 marks]**

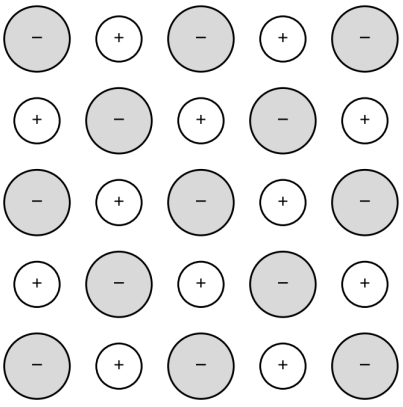
Complete the diagram to show the ions in magnesium oxide. Draw the outer electrons on each ion and write the charge in each box.



0 2 . 3 Write the formula of each ionic compound. **[5 marks]**

Positive ion	Negative ion	Formula of compound
Na^+	Cl^-	
Mg^{2+}	Cl^-	
Al^{3+}	O^{2-}	
Ca^{2+}	NO_3^-	
Fe^{3+}	OH^-	

0 3 The diagram shows part of the structure of sodium chloride.



0 3 . 1 What is the name given to this type of structure? **[1 mark]**

0 3 . 2 Describe the forces that hold the ions together in sodium chloride. **[2 marks]**

0 3 . 3 Sodium chloride has a high melting point (801 °C). **[3 marks]**
Explain why.

0 3 . 4 Solid sodium chloride does not conduct electricity. Molten sodium chloride does conduct electricity. **[3 marks]**
Explain why.

0 3 . 5 Magnesium oxide (melting point 2852 °C) has a much higher melting point than sodium chloride (801 °C). **[3 marks]**
Explain why.

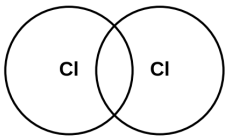
0 3 . 6 This type of diagram has limitations. **[2 marks]**
Give **two** limitations of this diagram of an ionic lattice.

0 4 This question is about covalent bonding.

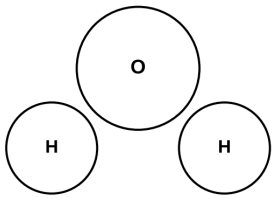
0 4 . 1 Complete the diagrams to show the outer electrons in a molecule of chlorine, water and methane. Use crosses (×) for electrons from one atom and dots (•) for electrons from the other atom. **[3 marks]**

Hydrogen has 1 outer electron, carbon has 4, oxygen has 6 and chlorine has 7.

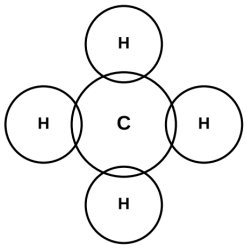
Cl_2



H_2O

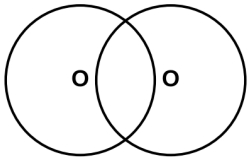


CH_4



0 4 . 2 What is a covalent bond? **[2 marks]**

0 4 . 3 An oxygen molecule (O_2) has a double bond. Complete the diagram to show the outer electrons in an oxygen molecule. **[2 marks]**



0 4 . 4 Methane has a low boiling point (−162 °C). **[3 marks]**

Explain why, in terms of its structure and bonding.

0 4 . 5 Explain why methane does not conduct electricity. **[2 marks]**

0 4 . 6 The table shows the boiling points of the first four alkanes. **[3 marks]**
Describe the trend and explain it.

Alkane	Formula	Boiling point in °C
methane	CH_4	−162
ethane	C_2H_6	−89
propane	C_3H_8	−42
butane	C_4H_{10}	−1

0 5 Poly(ethene) is a polymer. It is a solid at room temperature. Ethene (the monomer) is a gas at room temperature.

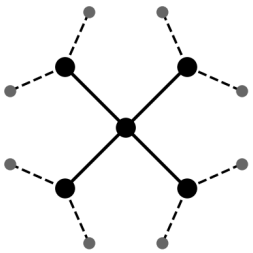
0 5 . 1 Explain why poly(ethene) is a solid at room temperature. **[3 marks]**

0 5 . 2 Describe the bonding in poly(ethene). **[2 marks]**

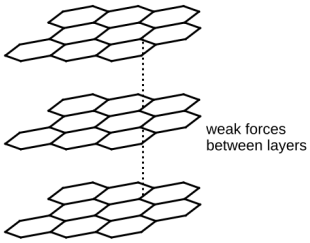
0 5 . 3 Poly(ethene) does not conduct electricity. Give a reason why. **[1 mark]**

0 6 The diagrams show the structures of some forms of carbon.

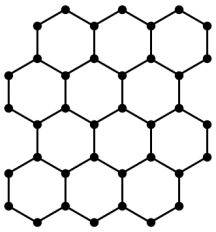
A



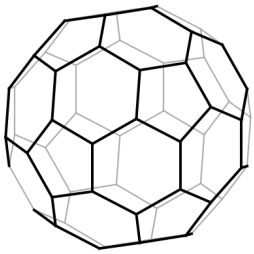
B



C



D



E



0 6 . 1 Complete the table. Write the letter of the structure that matches each name. **[5 marks]**

Name	**Letter**
diamond	
graphite	
graphene	
fullerene (C_{60})	
carbon nanotube	

0 6 . 2 Diamond has a very high melting point. **[3 marks]**
Explain why, in terms of its structure and bonding.

0 6 . 3 Diamond is very hard. Explain why. **[2 marks]**

0 6 . 4 Graphite conducts electricity. Diamond does not. **[3 marks]**
Explain why graphite conducts electricity.

0 6 . 5 Graphite is soft and slippery and is used as a lubricant. **[2 marks]**
Explain why.

0 6 . 6 Graphene is a single layer of graphite. **[3 marks]**
Give **two** properties of graphene and use them to suggest one use.

0 6 . 7 A fullerene molecule, such as C_{60}, can be used to carry a drug into the body. **[3 marks]**
Describe the structure of a fullerene and give **one** other use of fullerenes.

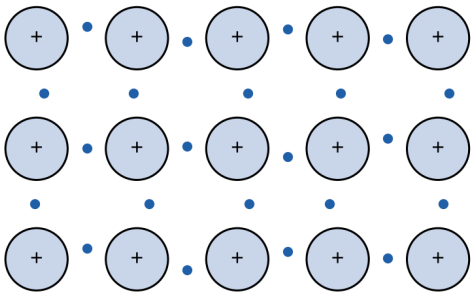
0 6 . 8 Carbon nanotubes are used to make strong, light sports equipment. **[2 marks]**
Suggest **two** properties of carbon nanotubes that make them suitable.

0 6 . 9 Silicon dioxide (silica) has a high melting point (1610 °C) and does not conduct electricity. **[2 marks]**
Explain why.

0 6 . 10 Graphite is used to make electrodes. **[2 marks]**
Give **two** properties of graphite that make it suitable.

0 7 This question is about metals and alloys.

0 7 . 1 The diagram shows the bonding in a metal. **[3 marks]**
Describe the structure and bonding of a metal.

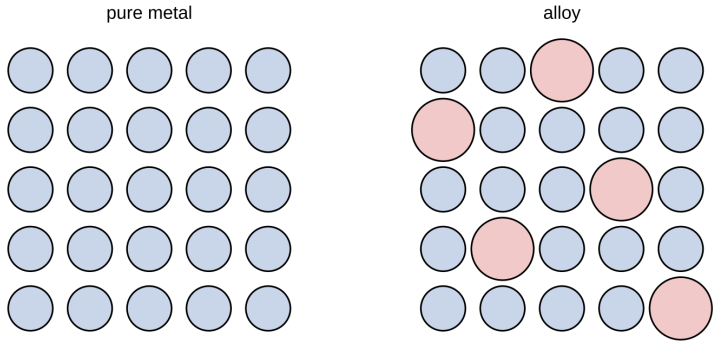


0 7 . 2 Explain why metals are good conductors of electricity. **[2 marks]**

0 7 . 3 Explain why metals have high melting points. **[2 marks]**

0 7 . 4 Pure metals can be bent and shaped. Explain why. **[2 marks]**

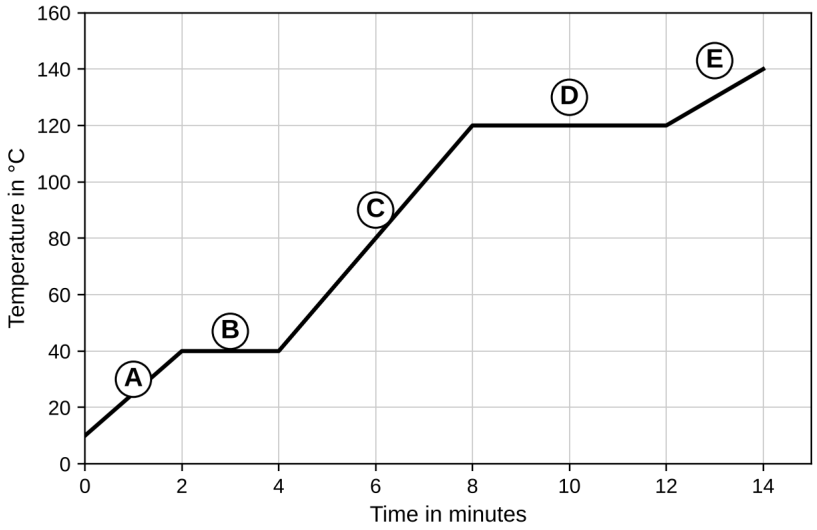
0 7 . 5 The diagrams show the particles in a pure metal and in an alloy. **[3 marks]**
Explain why the alloy is harder than the pure metal.



0 7 . 6 Pure gold is too soft to use for jewellery. Jewellery is often made from 18 carat gold, which is an alloy. **[2 marks]**
Explain why the alloy is better for making jewellery.

0 7 . 7 Name one alloy and give one use of it. **[2 marks]**

0 8 A student heats a solid, **Z**, until it becomes a gas. The graph shows how the temperature of **Z** changes with time.



0 8 . 1 Which two sections of the graph, **A** to **E**, show a change of state? **[2 marks]**

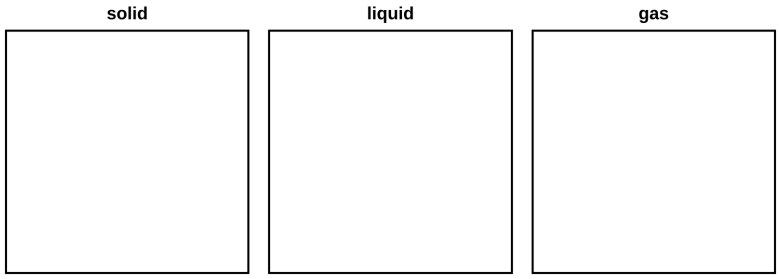
0 8 . 2 What are the melting point and the boiling point of **Z**? **[2 marks]**

0 8 . 3 What is the state of **Z** at 80 °C? **[1 mark]**

0 8 . 4 In section **B** the substance is still being heated but its temperature does not change. **[2 marks]**
Explain why.

0 9 This question is about the particle model.

0 9 . 1 Draw the arrangement of the particles in a solid, a liquid and a gas. Draw at least 12 particles in each box. **[3 marks]**



0 9 . 2 Complete the table to show the state of each substance at 20 °C. **[5 marks]**

Substance	Melting point in °C	Boiling point in °C	State at 20 °C
oxygen	−219	−183	
bromine	−7	59	
sodium chloride	801	1413	
mercury	−39	357	
ethanol	−114	78	

0 9 . 3 Magnesium reacts with hydrochloric acid. **[2 marks]**
Add the missing state symbols to the equation.

$Mg(____) + 2HCl(____) \rightarrow MgCl_2(____) + H_2(____)$

0 9 . 4 The particle model has limitations. Give **two** limitations of the simple particle model. **[2 marks]**

1 0 The table shows some properties of five substances, **P** to **T**. None of the letters are chemical symbols.

Substance	Melting point in °C	Conducts electricity as a solid	Conducts electricity when molten	Dissolves in water
P	801	no	yes	yes
Q	55	no	no	no
R	1085	yes	yes	no
S	1610	no	no	no
T	714	no	yes	yes

1 0 . 1 Which substance is a simple molecular substance? Write the letter. **[1 mark]**

1 0 . 2 Which **two** substances are ionic compounds? Write the letters. **[1 mark]**

1 0 . 3 Substance **R** is a metal. Give **two** pieces of evidence from the table. **[2 marks]**

1 0 . 4 Substance **S** has a giant covalent structure. **[3 marks]**
Explain why **S** has a high melting point and does not conduct electricity.

1 1 This question is about nanoparticles.

1 1 . 1 The table shows the diameters of different sizes of particle. **[2 marks]**
A particle has a diameter of 5×10^{-8} m. What type of particle is it? Give a reason.

Type of particle	Diameter in metres	Diameter in nanometres (nm)
coarse particles (dust)	1×10^{-5} to 2.5×10^{-6}	10 000 to 2500
fine particles	2.5×10^{-6} to 1×10^{-7}	2500 to 100
nanoparticles	1×10^{-7} to 1×10^{-9}	100 to 1

1 1 . 2 A cube has sides of length 100 nm. A second cube has sides of length 10 nm. **[4 marks]**
Calculate the surface area to volume ratio for each cube. (Surface area = 6 × side². Volume = side³.)

Answer = _____

1 1 . 3 Use your answers to describe what happens to the surface area to volume ratio when the side length of a cube decreases by a factor of 10. **[1 mark]**

1 1 . 4 Nanoparticles are used as catalysts. Explain why a catalyst made of nanoparticles is more effective than the same mass of larger particles. **[2 marks]**

1 1 . 5 Give **two** uses of nanoparticles. **[2 marks]**

1 1 . 6 Give **one** possible risk of using nanoparticles. **[1 mark]**

1 1 . 7 *Sun cream manufacturers use nanoparticles of titanium dioxide. Evaluate the use of nanoparticles in sun creams. **[4 marks]**

1 2 Stretch questions (grade 7–9).

1 2 . 1 *Compare the structures, bonding and properties of diamond and graphite. **[6 marks]**

1 2 . 2 *Sodium chloride (melting point 801 °C), methane (melting point −182 °C) and diamond (melting point about 3550 °C) have very different melting points.
Explain these differences in terms of structure and bonding. **[6 marks]**

END OF QUESTIONS

The Periodic Table of the Elements (elements 1–56)

1 **H** 1																	4 **He** 2
7 **Li** 3	9 **Be** 4											11 **B** 5	12 **C** 6	14 **N** 7	16 **O** 8	19 **F** 9	20 **Ne** 10
23 **Na** 11	24 **Mg** 12											27 **Al** 13	28 **Si** 14	31 **P** 15	32 **S** 16	35.5 **Cl** 17	40 **Ar** 18
39 **K** 19	40 **Ca** 20	45 **Sc** 21	48 **Ti** 22	51 **V** 23	52 **Cr** 24	55 **Mn** 25	56 **Fe** 26	59 **Co** 27	59 **Ni** 28	63.5 **Cu** 29	65 **Zn** 30	70 **Ga** 31	73 **Ge** 32	75 **As** 33	79 **Se** 34	80 **Br** 35	84 **Kr** 36
85.5 **Rb** 37	88 **Sr** 38	89 **Y** 39	91 **Zr** 40	93 **Nb** 41	96 **Mo** 42	[98] **Tc** 43	101 **Ru** 44	103 **Rh** 45	106 **Pd** 46	108 **Ag** 47	112 **Cd** 48	115 **In** 49	119 **Sn** 50	122 **Sb** 51	128 **Te** 52	127 **I** 53	131 **Xe** 54
133 **Cs** 55	137 **Ba** 56	57–71															

Group 1 → 18: left to right. Periods: top to bottom.

Relative atomic mass is shown above each symbol and atomic (proton) number below. Elements 57–71 are not shown.

MARK SCHEME – Topic 4.2: Bonding, structure and properties of matter

How to use this mark scheme: mark your own work in a different colour. A semicolon (;) separates marking points and each is worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed to gain the mark. **Allow** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. Marks are for the *science*, not the wording.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	sodium chloride: ionic ; carbon dioxide: covalent ; copper: metallic ; magnesium oxide: ionic ; methane: covalent ; iron: metallic ;	6 rows correct = 3 marks; 4 or 5 correct = 2 marks; 2 or 3 correct = 1 mark.	3	AO1 4.2.1.1
01.2	ionic ; covalent ;		2	AO1 4.2.1.1
02.1	a sodium atom loses / transfers one electron ; a chlorine atom gains that electron ; to form Na^+ (2,8) and Cl^- (2,8,8) ions with full outer shells ; oppositely charged ions attract (strong electrostatic attraction = ionic bond) ;	Common error: saying electrons are 'shared'. In ionic bonding electrons are **transferred**.	4	AO1 4.2.1.2
02.2	magnesium ion: charge 2+ (outer shell empty or showing 8 electrons) ; oxide ion: 8 outer electrons, 2 dots and 6 crosses ; oxide ion charge 2− (both ions in brackets) ;	Allow Mg^{2+} with no electrons or with 8 electrons in the outer shell. Do not accept charges written without + or −.	3	AO2 4.2.1.2
02.3	NaCl ; $MgCl_2$; Al_2O_3 ; $Ca(NO_3)_2$; $Fe(OH)_3$;	One mark each. Method: balance the charges. Al^{3+} × 2 = 6+, O^{2-} × 3 = 6−. Brackets are needed around NO_3 and OH when there is more than one.	5	AO2 4.2.1.2
03.1	giant ionic lattice	Allow 'giant lattice' or 'ionic lattice'.	1	AO1 4.2.2.1
03.2	strong electrostatic forces of attraction ; between oppositely charged ions (in all directions) ;	Do not accept 'strong covalent bonds' or 'intermolecular forces'.	2	AO1 4.2.2.1
03.3	it has a giant lattice of ions ; with strong electrostatic forces between oppositely charged ions ; a lot of energy is needed to overcome (break) these forces ;	Do not accept 'covalent bonds are broken' or 'molecules'.	3	AO1 4.2.2.1
03.4	in the solid the ions are in fixed positions / cannot move ; when molten the ions are free to move ; and carry the charge (through the liquid) ;	Allow the same explanation for sodium chloride dissolved in water. Do not accept 'electrons carry the charge' - in ionic compounds it is the ions.	3	AO2 4.2.2.1
03.5	Mg^{2+} and O^{2-} ions have higher charges than Na^+ and Cl^- ; so stronger electrostatic forces of attraction between the ions ; more energy is needed to overcome the forces ;	Grade 8–9 reasoning: compare the **charges** on the ions.	3	AO3 4.2.2.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.6	it is two-dimensional / does not show the 3D arrangement ; does not show the sizes of the ions correctly ; does not show the forces between ions ; shows only a small part of the lattice (any two) ;	Any two.	2	AO3 4.2.2.1
04.1	Cl_2: one shared pair in the overlap and 3 non-bonding pairs on each chlorine (8 outer electrons each) ; H_2O: two shared pairs, two non-bonding pairs on the oxygen ; CH_4: four shared pairs between carbon and the four hydrogens ;	One mark for each molecule correct. Each shared pair must be in the overlap of the circles.	3	AO2 4.2.1.3
04.2	a shared pair of electrons ; between two (non-metal) atoms ;	Allow 'strong' as an extra detail but it does not gain a mark alone.	2	AO1 4.2.1.3
04.3	two shared pairs (4 electrons) in the overlap ; each oxygen atom has two other non-bonding pairs (8 outer electrons in total) ;	Crosses and dots must be used to show where the electrons come from.	2	AO2 4.2.1.3
04.4	methane is a small molecule (simple molecular substance) ; there are weak forces between the molecules (intermolecular forces) ; little energy is needed to overcome these forces (the covalent bonds do not break) ;	**Most common lost mark:** saying covalent bonds break when a substance boils. They do not.	3	AO1 4.2.2.2
04.5	no ions / no delocalised electrons ; so there are no charged particles free to move / molecules have no overall charge ;		2	AO1 4.2.2.2
04.6	boiling point increases as the molecules get bigger / as the number of carbon atoms increases ; intermolecular forces are stronger between larger molecules ; so more energy is needed to overcome them ;	Boiling points rise because intermolecular forces get stronger, not because covalent bonds are stronger.	3	AO3 4.2.2.2
05.1	polymers have very large molecules (long chains) ; so the intermolecular forces between the molecules are stronger ; more energy is needed to overcome them (so the melting point is above room temperature) ;	Do not accept 'the covalent bonds are stronger'.	3	AO2 4.2.2.2
05.2	atoms in each chain are joined by strong covalent bonds ; chains are held together by (weaker) intermolecular forces ;		2	AO1 4.2.2.2
05.3	no free electrons / no ions / no charged particles that are free to move		1	AO1 4.2.2.2
06.1	diamond: A ; graphite: B ; graphene: C ; fullerene (buckminsterfullerene, C_{60}): D ; carbon nanotube: E ;		5	AO1 4.2.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.2	giant covalent structure ; many strong covalent bonds (each carbon bonded to four others) ; a lot of energy is needed to break these bonds ;	Do not accept 'intermolecular forces' or 'ionic'.	3	AO1 4.2.3.1
06.3	each carbon atom forms four covalent bonds ; in a rigid giant structure / bonds are strong in all directions ;		2	AO1 4.2.3.1
06.4	each carbon atom forms three covalent bonds ; so one electron from each carbon is delocalised (free to move) ; the delocalised electrons carry charge through the structure ;	Diamond has no free electrons (all four outer electrons are in bonds).	3	AO2 4.2.3.1
06.5	graphite has layers of carbon atoms ; with weak forces between the layers so the layers slide over each other ;	Do not accept 'weak covalent bonds between layers'. The forces between layers are weak intermolecular-type forces.	2	AO2 4.2.3.1
06.6	conducts electricity / very strong / very thin / transparent / flexible / high melting point (any two properties) ; (second property) ; use linked to a property, e.g. electronics, composite materials, touch screens ;	One mark for each property (two marks) and one mark for a matching use.	3	AO2 4.2.3.2
06.7	hollow ; shape made of rings of carbon atoms (hexagons and some pentagons) ; other use: lubricant / catalyst (support) ;	Allow 'cage' or 'ball' for hollow.	3	AO1 4.2.3.2
06.8	very high tensile strength ; low density / light ; conduct electricity (any two) ;	Any two.	2	AO2 4.2.3.2
06.9	giant covalent structure with strong covalent bonds (a lot of energy to break) ; no free electrons / no ions so no charge can flow ;		2	AO2 4.2.2.3
06.10	conducts electricity ; high melting point / unreactive / inert ;	Any two.	2	AO2 4.2.3.1
07.1	a regular arrangement (lattice) of positive metal ions ; surrounded by a sea of delocalised electrons ; strong electrostatic attraction between the positive ions and the delocalised electrons ;	Do not accept 'shared electrons' - that is covalent bonding.	3	AO1 4.2.1.4
07.2	delocalised electrons are free to move ; and carry charge through the metal ;		2	AO1 4.2.2.4
07.3	strong electrostatic attraction between positive ions and delocalised electrons / strong metallic bonding ; a lot of energy needed to overcome the attraction ;		2	AO1 4.2.2.4
07.4	the layers of atoms / ions can slide over each other ; the metallic bonding is not broken / the electrons hold the layers together ;		2	AO2 4.2.2.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.5	the alloy contains atoms of different sizes ; which distort the regular layers ; so the layers cannot slide over each other as easily ;	Do not accept 'the alloy has stronger metallic bonds'.	3	AO2 4.2.2.5
07.6	alloy is harder / stronger than pure gold ; so it keeps its shape / does not wear or bend easily ;		2	AO3 4.2.2.5
07.7	steel / brass / bronze / solder / duralumin / amalgam ... ; a correct use (e.g. steel – buildings or cars; brass – taps or instruments; solder – joining wires) ;		2	AO1 4.2.2.5
08.1	B ; D ;		2	AO1 4.2.2.6
08.2	melting point 40 °C ; boiling point 120 °C ;	Units are needed; allow without ° if C is written.	2	AO2 4.2.2.6
08.3	liquid		1	AO2 4.2.2.6
08.4	the energy is used to overcome the forces between the particles / break the bonds between particles ; so the substance melts (changes state) instead of the particles speeding up / temperature rising ;	Do not accept 'the covalent bonds break' for a simple molecular substance.	2	AO2 4.2.2.6
09.1	solid: regular pattern, touching / very close ; liquid: random arrangement, touching / close together ; gas: far apart, random ;	One mark for each box. Solid must be in a regular pattern; liquid not in a pattern.	3	AO1 4.2.2.6
09.2	oxygen: gas ; bromine: liquid ; sodium chloride: solid ; mercury: liquid ; ethanol: liquid ;	If the melting point is below 20 °C and the boiling point is above 20 °C the substance is a liquid.	5	AO2 4.2.2.6
09.3	s, aq, aq, g ; all four correct for 2 marks ;	4 correct = 2 marks; 2 or 3 correct = 1 mark. (aq) means dissolved in water.	2	AO1 4.2.1.1
09.4	there are no forces between the particles shown ; the particles are shown as solid spheres ; the spaces between particles are not to scale / particles are not really spheres (any two) ;	Any two.	2	AO3 4.2.2.6
10.1	Q	Low melting point, does not conduct, insoluble.	1	AO3 4.2.2
10.2	P and T (both needed)	High melting point, conduct only when molten (or dissolved), dissolve in water.	1	AO3 4.2.2.1
10.3	conducts electricity when solid ; high melting point ; conducts when molten (any two) ;	Any two. Do not accept 'does not dissolve' on its own.	2	AO3 4.2.2.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
10.4	strong covalent bonds ; a lot of energy needed to break the bonds in the giant structure ; no free electrons / ions (all outer electrons are in bonds) ;		3	AO2 4.2.2.3
11.1	nanoparticle ; 5×10^{-8} m (50 nm) is between 1 nm (1×10^{-9} m) and 100 nm (1×10^{-7} m) ;		2	AO2 4.2.4.1
11.2	100 nm cube: surface area = 6 × 100 × 100 = 60 000 nm² and volume = 1 000 000 nm³ ; ratio = 0.06 : 1 ; 10 nm cube: surface area = 600 nm² and volume = 1000 nm³ ; ratio = 0.6 : 1 ;	Allow 0.06 and 0.6 (per nm) with no units. Shortcut: ratio = 6 ÷ side length.	4	AO2 4.2.4.1
11.3	the ratio increases by a factor of 10		1	AO2 4.2.4.1
11.4	nanoparticles have a much larger surface area to volume ratio ; so more of the catalyst is exposed to the reactants / more collisions at the surface ;		2	AO2 4.2.4.2
11.5	sun creams ; deodorants / cosmetics ; medicine (drug delivery) ; catalysts ; electronics / fabrics / coatings (any two) ;	Any two.	2	AO1 4.2.4.2
11.6	may be toxic / harmful to health or the environment ; can enter the body / cells ; effects not fully known (any one) ;	Any one.	1	AO1 4.2.4.2
11.7	**Level 2 (3–4 marks)**: A clear answer covering the key ideas with correct reasoning linking them. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • benefits: effective at blocking UV light / transparent so no white marks / only a small amount needed • risks: nanoparticles may be absorbed through the skin or enter cells / long-term effects not yet known • environmental risk when washed into water systems • a conclusion that weighs benefits against risks (e.g. more testing needed)		4	AO3 4.2.4.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
12.1	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • both are giant covalent structures made only of carbon, with very high melting points • diamond: each carbon bonded to four others, rigid, very hard • graphite: each carbon bonded to three others in layers of hexagons • graphite has one delocalised electron per atom so it conducts electricity; diamond has no free electrons so it does not • graphite layers held by weak forces so they slide and graphite is soft / slippery • diamond is hard because its bonds are strong in all directions		6	AO1 4.2.3.1
12.2	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • sodium chloride: giant ionic lattice; strong electrostatic forces between oppositely charged ions; high mp • methane: small molecules; weak intermolecular forces; little energy needed to overcome them; very low mp (covalent bonds do not break) • diamond: giant covalent structure; very many strong covalent bonds; a huge amount of energy to break them; highest mp • melting point depends on the strength of the forces being overcome • ionic and covalent bonds are both strong but different amounts of bonding must be broken • link: energy needed increases from weak intermolecular forces < ionic lattice < giant covalent network		6	AO1 4.2.2

Where marks are lost in Topic 4.2

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Boiling or melting a simple molecular substance (methane, water, chlorine) breaks intermolecular forces**, not covalent bonds. This is the single most common lost mark in this topic.
- ☐ **Ionic compounds are not molecules.** Use the words ions, giant lattice and electrostatic attraction. Never say 'sodium chloride molecules'.
- ☐ **Conductivity needs two ideas:** charged particles (ions or electrons) *and* free to move. Solid ionic: ions are fixed, so no conduction.
- ☐ **Delocalised electrons** are in metals and graphite only. Diamond, silica and molecules have none.
- ☐ **Ionic bonding = transfer** of electrons. **Covalent = sharing.** Metallic = sea of delocalised electrons. Do not mix the verbs.
- ☐ **Comparing melting points of ionic compounds:** compare the charges (and sizes) of the ions. $Mg^{2+}O^{2-}$ beats $Na^{+}Cl^{-}$ because 2+/2− attract more strongly than 1+/1−.
- ☐ **Alloys:** different sized atoms distort the layers so they cannot slide. Do not say the bonds are stronger.
- ☐ **Graphite:** three bonds per carbon, one delocalised electron per carbon, weak forces between layers. Write all three.
- ☐ **Dot-and-cross diagrams:** draw all outer electrons, put ions in square brackets, include the charge, and make shared pairs sit in the overlap.
- ☐ **Nanoparticle calculations:** surface area = 6 × side² and volume = side³, so the ratio is 6 ÷ side. Smaller cube means bigger ratio.
- ☐ **Compare questions:** use 'whereas' or 'but' in every sentence so each point about one structure is matched with the other.